



Technical Service Bulletin: TSB190107

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Copper Sealing Ring Replacement on Injector: Engine Power Output Low and Engine Runs Rough or Misfires

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK78 CM500
- QSK78 CM850
- QSK78 CM2250 K104
- QSK78 CM2350 K126
- QSK78 CM2350 K137
- QSK78 PG CM500
- QSK78 PG CM850
- QSK45 CM500
- QSK60 CM500
- QSK60 CM850
- QSK60 CM2250
- QSK60 CM2350

Issue

Symptom:

- Engine runs rough or misfire
- Engine power output low/ Low EGT Fault Code(s) 661-668, 671-678, 711-718, or 721-728
- Engine black smoke excessive

Root Cause:

- Corrosion and stress cracks found on copper injector sealing rings. Issue is more prominent at altitude.

Resolution

The existing copper injector sealing ring is being replaced by a new stainless-steel injector sealing ring.

Service Instructions

- Replace existing copper injector sealing ring with new stainless-steel injector sealing ring.
- See corresponding Service Manual. Reference Procedure 006-026 in Section 6. Remove injector sealing ring by prying one hold tab over the injector nozzle rim at a time.
- The stainless-steel sealing rings are **only** available in standard sizes. If oversized sealing rings are required these are **only** available in copper.

Service Parts Availability

Service parts are available. See Table 1 below for part numbers.

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
QSK78 Injector Seal	4307148	No	No	5462083
QSK45/60 Injector seal	3867687	Yes	Yes	5462083
QSK60 Injector Seal	3867687	Yes	Yes	5462083
QSK60 M Injector Seal	4307148	Yes	Yes	5462083

Part Identification

- New injector sealing ring is stainless-steel.
- Existing injector sealing ring is copper.

Part Inventory Action

Copper sealing rings in stock and that have been removed are to be discarded.

Production Status

Implemented for production. See Table 2 below.

Table 2, Production Information		
ESN First	Build Date ¹	Plant
66305406 (QSK78)	1-Jul-19	Daventry
33219763 (QSK45/60)	6-Sep-19	Daventry
¹ Engine build date can be found on engine dataplate.		

Document History

Date	Details
2019-6-20	Module Created
2019-10-7	Non-Product Problem Solving (PPS)
2020-2-4	Updated table 1.

Last Modified: 05-Feb-2020
